

HARRINGTON, PE, Canada

WMO: 713500

Lat: 46.3436N

Lon: 63.1697W

Elev: 174

StdP: 14.60

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.7	2.4	-10.2	3.2	-0.5	-6.0	4.0	3.9	29.7	23.1	26.9	25.2	9.5	280	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	80.1	68.9	77.8	67.3	75.4	66.0	71.2	77.4	69.7	75.3	68.1	73.3	10.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	107.2	75.2	67.6	101.9	73.4	66.0	96.4	71.6	35.2	77.6	33.8	75.3	32.6	73.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.8	19.5	-6.8	85.0	3.4	2.5	-9.2	86.8	-11.2	88.3	-13.1	89.7	-15.5	91.4		
(5)			-7.4	73.8	3.4	1.3	-9.9	74.7	-11.9	75.4	-13.8	76.2	-16.3	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	21.1	20.6	26.9	38.1	48.3	57.8	66.5	66.2	58.9	48.7	38.5	27.8		
	DBStd	17.80	10.37	9.29	8.01	7.08	7.01	6.54	4.84	4.49	6.36	6.87	8.08	9.09		
	HDD50	4085	895	823	716	366	118	10	0	0	5	109	356	687		
	HDD65	8076	1360	1243	1180	808	517	229	41	41	201	508	796	1152		
	CDD50	1681	0	0	1	8	66	244	511	502	272	67	10	0		
	CDD65	197	0	0	0	0	1	13	87	77	18	1	0	0		
	CDH74	804	0	0	0	0	13	59	380	297	54	1	0	0		
	CDH80	74	0	0	0	0	0	3	40	24	7	0	0	0		
	WSAvg	9.5	10.5	10.0	10.7	10.3	9.7	8.8	7.9	7.7	8.8	9.3	10.2	10.5		
	PrecAvg	47.9	4.4	3.7	3.9	3.6	3.7	3.7	3.3	3.5	3.8	4.4	4.7	5.2		
(15) Precipitation	PrecMax	62.2	10.4	6.9	7.5	7.0	8.6	8.0	9.1	8.3	7.0	8.5	8.5	10.2		
	PrecMin	33.4	1.3	1.1	1.8	0.6	1.0	0.8	0.4	0.4	0.9	1.1	1.6	2.5		
	PrecStd	6.8	1.6	1.4	1.3	1.4	1.7	1.7	1.9	1.9	1.5	1.9	1.5	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	46.7	53.3	64.8	75.8	79.5	84.0	82.8	79.2	70.3	63.0	54.4		
		MCWB	51.0	44.5	44.5	56.3	61.6	66.0	71.3	70.3	69.6	63.0	59.9	53.0		
	2%	DB	43.8	41.2	44.9	59.2	69.3	75.5	80.6	79.8	74.8	65.5	58.9	48.8		
		MCWB	42.8	39.4	42.1	53.3	59.0	64.2	69.6	68.5	66.5	60.2	56.9	47.5		
	5%	DB	39.5	36.9	40.9	53.9	64.4	72.5	78.4	77.3	71.8	62.3	55.1	44.2		
		MCWB	37.9	34.8	37.6	49.1	55.5	63.1	68.0	67.1	65.5	58.3	52.6	42.9		
	10%	DB	35.1	33.6	37.4	49.7	60.3	69.1	75.6	74.9	69.0	59.3	51.2	40.0		
		MCWB	33.7	31.9	34.5	44.8	53.6	61.3	66.6	66.0	63.7	55.2	48.7	38.3		
	0.4%	WB	50.8	44.9	46.9	59.3	64.6	70.3	73.3	72.9	71.0	65.8	60.5	53.1		
		MCDB	52.3	45.9	50.1	63.7	71.7	76.2	80.7	79.5	76.6	68.5	62.5	54.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.0	39.7	42.5	54.6	60.8	67.4	71.6	71.0	68.4	61.4	57.1	47.6		
		MCDB	43.8	41.0	44.5	57.6	66.0	72.4	78.4	76.7	72.5	64.6	59.0	48.6		
	5%	WB	38.1	35.2	38.2	49.9	57.7	64.9	70.0	69.4	66.6	58.8	53.2	43.0		
		MCDB	39.2	36.5	40.5	53.0	62.6	69.9	75.7	75.1	70.6	61.8	54.7	44.1		
	10%	WB	33.9	32.1	34.5	45.4	54.3	62.0	68.4	67.7	64.4	55.7	48.7	38.8		
		MCDB	34.9	33.5	37.1	49.3	59.4	67.6	73.7	73.2	68.2	58.8	50.9	40.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	12.8	11.9	13.2	15.3	14.7	14.2	14.1	13.9	11.9	11.3	11.0		
		MCDBR	19.4	16.7	16.3	20.7	22.9	19.9	18.3	17.4	16.7	15.2	16.9	16.8		
		MCWBR	19.8	15.8	14.3	15.4	14.5	11.7	9.9	9.0	10.9	12.1	16.1	17.5		
	5% WB	MCDBR	19.3	17.3	16.1	18.8	19.9	16.9	15.5	14.8	14.7	14.7	16.8	16.5		
		MCWBR	19.9	16.9	15.5	16.2	14.5	11.6	9.5	9.0	11.1	13.2	16.8	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.288	0.318	0.347	0.365	0.381	0.406	0.393	0.355	0.333	0.312	0.290		
	taud		2.339	2.304	2.320	2.399	2.417	2.388	2.354	2.388	2.490	2.526	2.506	2.464		
	Ebn at Noon		251	281	288	289	286	281	272	271	273	264	248	246		
	Edh at Noon		24	30	34	35	36	37	38	35	29	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		337	560	993	1330	1618	1761	1790	1578	1206	744	384	260		
	RadStd		22	76	95	121	147	133	120	99	105	76	47	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	

Nomenclature: See separate page